A Reverse-Engineered College Basketball Rating Model

A KenPom-Inspired Algorithm

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Executive summary. This report documents a college basketball rating system that I built to approximate the predictive logic behind Ken Pomeroy's KenPom system while still remaining fully tunable. The model converts box score data into possession-based efficiencies, caps blowout margins, adjusts for home court advantage, rescales performances by opponent strength, and weights recent games more heavily than older ones. The resulting rankings are directionally similar to KenPom's current board while still expressing a few meaningful differences. Those differences are exactly the point: by building my own system, I can inspect every assumption, test alternatives, and understand why the model likes or dislikes a team.

Introduction / Motivation

In the NCAA, there are 361 Division 1 schools, all of which belong to a conference of around 8-18 schools. Each school plays about 30-35 games per season, around 12 of which are against teams outside of their conference. At the end of each season, the NCAA March Madness Tournament consists of a 64-team, single elimination bracket and crowns a singular winner. Out of the 63 total games in the tournament, there are few matchups between two teams that have already faced each other earlier in the year.

It is because of this that analytical metrics like KenPom exist. KenPom is system created by Ken Pomeroy that aims to quantify how well each team has performed, adjusting for quality of opponents. KenPom is great because, by adjusting for strength of schedule, we can compare two teams that have never played against each other, and even have minimal common opponents. However, Pomerow keeps his algorithm behind a paywall, so I decided to build my own.

I wanted to build my own version of a KenPom-style rating system for two reasons. First, I wanted a predictive metric that reflects the way modern basketball analysts think about team strength: not just wins and losses, but efficiency, pace, opponent quality, game location, and recent form. Second, I wanted a model that I could tweak myself. Public systems such as KenPom are extremely useful, but the exact formula is proprietary. Rebuilding a similar framework from scratch forces me to think carefully about what each modeling choice is supposed to measure.

KenPom has described his own ratings as a *predictive* system meant to estimate how strong a team would be "if it played tonight". He also emphasizes adjusted efficiency margin - offensive efficiency minus defensive efficiency - as a clean measure of team strength over 100 possessions.[1] That is the exact spirit of this project. My goal is not to duplicate the private formula perfectly. My goal is to build a transparent approximation that behaves like a serious predictive model and can be improved over time.

[1] Ken Pomeroy, "Ratings Explanation," KenPom blog, Nov. 29, 2006; and "Ratings Methodology Update," Oct. 4, 2016.

The Algorithm

Assumptions

When designing this system, I made many basketball logic-related assumptions that are reflected in the algorithm. The overarching principle can be broken down into two parts:

- Binary wins and losses matter much less than general performance. A close win versus a really bad team is much worse than a close loss versus a really good team.

Although it might be the most important, this isn't the only assumption that was made:

- Possession based efficiencies are much more useful points scored/allowed. A team can easily score more points if their game had many more possessions, but in a theoretical head to head matchup, both teams will have the same amount possessions with and without the ball.
- Blowouts are pointless. For example, outscoring a team by 40 points has the same result as only winning by 30.
- Home court advantage affects both offensive and defensive efficiencies. For example, a team that had an offensive efficiency of 90 at home, did worse than a team with the same offensive efficiency on the road.
- Lastly, recently played games are more significant than those played a while ago.

What my metric is trying to measure

At a high level, the metric estimates how many points per 100 possessions a team would score and allow against the average Division I opponent on a neutral floor. The final rating is:

$$\text{Adjusted Net Rating} = \text{Adjusted Offensive Efficiency} - \text{Adjusted Defensive Efficiency}.$$

This mirrors KenPom's adjusted efficiency margin framework. In his system, AdjEM is the number of points a team would be expected to outscore an average Division I opponent by over 100 possessions.[2] My model uses the same target quantity because it is both predictive and interpretable.

Data collection pipeline

The raw dataset was built by scraping the ncaa.com website, which allowed me to map each game to its calendar date from the daily scoreboard feed, and then pull team-level box score statistics for every completed game. For each team-game observation, the scraper stores the fields needed for an efficiency model: shooting volume, offensive rebounds, turnovers, free throws, made threes, made field goals, and made free throws.

This is an important design choice. The model is based entirely on reproducible box score data rather than subjective information. That makes the pipeline auditable end to end: from API call, to raw CSV, to adjusted ratings.

[2] Ken Pomeroy, "Ratings Methodology Update," KenPom blog, Oct. 4, 2016.

Possessions and raw scoring output

The algorithm starts from a standard possession estimate:

$$\text{Possessions} = \text{FGA} - \text{ORB} + \text{TO} + 0.475 \times \text{FTA}.$$

This is a common college basketball approximation and is consistent with possession-based efficiency analysis. Points scored are reconstructed from the box score as:

$$\text{Points Scored} = 3 \times \text{3PM} + 2 \times (\text{FGM} - \text{3PM}) + \text{FTM}.$$

The purpose of these two steps is straightforward: remove pace from the evaluation. A team should not look elite merely because it plays quickly. By converting performance to a per-100-possession basis, the metric compares teams on the quality of their possessions rather than the quantity of them.

Margin capping

One of the most important nuances in the formula is margin capping. Rather than using the actual score margin directly, the algorithm clips it to a symmetric threshold:

$$\text{Margin}_{\text{capped}} = \min\{C, \max[-C, \text{Margin}]\},$$

where C is a tunable margin cap. After tuning, the best value in this project was $C = 12$.

Why cap margin at all? Because predictive models should reward dominance, but they should not let a handful of extreme blowouts overpower the signal from the rest of the schedule. KenPom has written that scoring margin does contain predictive information, including at larger margins, but that the relationship is not simple and must be handled carefully.[3] My compromise is to preserve the direction and general strength of a result while limiting the ability of a couple runaway games to distort the season-long profile.

Once the margin is capped, the total points in the game are redistributed into capped scored and allowed values:

$$\text{Points Scored}_{\text{capped}} = \frac{\text{Total Points} + \text{Margin}_{\text{capped}}}{2},$$
$$\text{Points Allowed}_{\text{capped}} = \frac{\text{Total Points} - \text{Margin}_{\text{capped}}}{2}.$$

Those values are then converted into offensive and defensive efficiencies:

$$\text{Offensive Efficiency} = 100 \times \frac{\text{Points Scored}_{\text{capped}}}{\text{Possessions}},$$

[3] Ken Pomeroy, "Evidence that scoring margin matters," KenPom blog, Jan. 22, 2013.

$$\text{Defensive Efficiency} = 100 \times \frac{\text{Points Allowed}_{\text{capped}}}{\text{Possessions}}.$$

This is a strong modeling choice because it treats a 30-point win as clearly better than a 5-point win, but not 6 times as informative.

Home-court adjustment

After the initial efficiency calculation, the notebook applies a fixed home-court adjustment by shifting the home team's offense and defense in opposite directions, and doing the reverse for the road team. The tuned home-court value is $H = 2.0$ points per 100 possessions:

$$\text{Home Offensive Efficiency} \leftarrow \text{Home Offensive Efficiency} - H,$$

$$\text{Home Defensive Efficiency} \leftarrow \text{Home Defensive Efficiency} + H,$$

$$\text{Away Offensive Efficiency} \leftarrow \text{Away Offensive Efficiency} + H,$$

$$\text{Away Defensive Efficiency} \leftarrow \text{Away Defensive Efficiency} - H.$$

The reasoning is that location affects observed efficiency. If a home team posts a certain raw efficiency, part of that performance should be attributed to venue advantage rather than pure team quality. KenPom has written extensively about home-court advantage and the difficulty of estimating it precisely, noting that it is real but noisy and generally smaller than many fans assume.[4] My model follows that same logic with a simpler constant adjustment.

Opponent adjustments

Raw efficiency is still incomplete because it ignores who the team played. Scoring 112 points per 100 possessions against an elite defense is more impressive than scoring the same number against a weak one. To fix that, the model rescales each game's efficiency by the opponent's current estimated strength relative to the national average efficiency:

$$\text{AdjOff}_g(A) = \text{OffEff}_g(A) \times \frac{\text{National Avg}}{\text{Opp Current AdjDef}},$$

$$\text{AdjDef}_g(A) = \text{DefEff}_g(A) \times \frac{\text{National Avg}}{\text{Opp Current AdjOff}}.$$

This is the heart of the algorithm. It creates an iterative, opponent-aware rating process. Strong performances against strong competition get amplified. Strong-looking box score lines against weak teams get pulled back toward the mean.

Two implementation details are especially important here:

- If a team has no prior history, the model falls back to the national average efficiency. This prevents the early-season ratings from exploding due to missing priors.
- For a team's first game, the notebook keeps the raw efficiency rather than forcing a questionable

[4] Ken Pomeroy, "Mining point spread data: home court advantage," KenPom blog, May 21, 2018.

opponent adjustment off almost no information. This is a conservative choice that improves stability.

Recency weighting

The final major nuance is recency weighting. Instead of taking a plain average of all adjusted efficiencies, the notebook uses an exponentially decaying weighted average. If a team's kth most recent game is indexed by k, their weights are:

$$w_k = 0.5^{k/\text{half-life}}.$$

The best half-life found by tuning was 20 games. Intuitively, that means a game from roughly 20 contests ago receives about half the weight of the most recent one.

This is one of the clearest places where I intentionally leaned into the predictive framing. A team in March is not the same team it was in November. Rotations tighten, freshmen develop, injured players return, and coaching adjustments accumulate. KenPom's public explanations also frame the ratings as a "play tonight" system rather than a full-season achievement score.[5] My recency weighting is a direct attempt to encode that idea.

Hyperparameter tuning

To avoid choosing the key knobs by feel alone, I evaluated combinations of the margin cap, home-court advantage, and half-life against out-of-sample style game predictions. For games from Jan. 1, 2026 onward (to avoid early-season sample variation), I formed a pregame rating for each team and predicted home margin as:

$$\widehat{\text{Home Margin}} = \text{Home AdjNet}_{\text{pre}} - \text{Away AdjNet}_{\text{pre}} + H.$$

The score used for tuning is mean absolute error between predicted and actual home margin. The best parameter set found in the notebook is shown below.

Configuration	Margin cap	Home-court adj.	Half-life
Initial trial	16	3.75	12
Best tuned model	12	2.0	20

Table 1: Model tuning results. The final report uses the tuned configuration.

This tuning step is important because it turns the project from a hand-built formula into a tested predictive model. It also gives every modeling choice a measurable consequence.

Results

Top of the board

Using the tuned settings $(C, H, \text{half-life}) = (12, 2, 20)$, the model's top 15 teams were as follows:

[5] Ken Pomeroy, "Ratings Explanation," KenPom blog, Nov. 29, 2006.

Rank	Team	Adj. Off.	Adj. Def.	Adj. Net
1	Duke	121.36	97.53	23.83
2	Arizona	120.50	96.84	23.66
3	Michigan	119.54	96.72	22.82
4	Purdue	126.25	104.97	21.28
5	Houston	118.23	97.42	20.80
6	St. John's (NY)	115.39	95.48	19.91
7	Illinois	123.66	103.99	19.67
8	Gonzaga	116.74	97.32	19.42
9	Saint Mary's (CA)	120.16	100.78	19.38
10	UConn	117.29	97.91	19.38
11	Iowa St.	117.13	97.98	19.15
12	Virginia	118.19	99.33	18.87
13	Florida	118.63	99.95	18.68
14	Michigan St.	118.06	99.89	18.17
15	Texas Tech	121.05	103.40	17.65

Table 2: Top 15 teams in the custom rating model.

At the top, the model passes a basic sanity check: the same general group of contenders appears near the top as in KenPom's ratings page. As of games through all conferense tournaments, KenPom's top 10 were Michigan, Arizona, Duke, Houston, Florida, Illinois, Iowa State, Purdue, Michigan State, and Connecticut.[6] My model also places Duke, Arizona, Michigan, Purdue, Houston, Illinois, Iowa State, Michigan State, and UConn in the upper tier, which is a strong sign that the architecture is capturing real signal rather than producing arbitrary outputs.

To make that comparison more concrete, 18 of my top 25 teams also appeared in KenPom's top 25, and the Spearman rank-order correlation coefficient on those shared teams is approximately 0.71. That is not a perfect match, but it is exactly what I would expect from an independent model: broad overall agreement with a few meaningful deviations driven by different assumptions.

Where my model differs

The most interesting part of the project is not where the model agrees with KenPom. It is where it disagrees and gives me something to investigate. As you will see, the recency-weighted approach of my model is responsible for a lot of these discrepancies.

These gaps are useful because they reveal what the formula is emphasizing. A higher Saint Mary's placement, for example, suggests my opponent adjustment and recency weighting were more willing to buy into the Gaels' based on their underlying efficiency profile. A lower Florida placement suggests my model was less persuaded by the Gators' season-long dominance once the capped margins and weighting scheme were applied.

Tournament read-through: where the model has looked better and worse

The early 2026 NCAA tournament provides a helpful reality check. Through games completed before the Sweet 16, there are already examples where my model looked worse than KenPom and examples where it looked more reasonable. Before we get into how my model has stacked up

[6] KenPom ratings page, data through games of Sunday, March 22, 2026.

against KenPom in the NCAA tournament, it's important to note the high-variance nature of the tournament. The single elimination format allows the worse team to become victorious more often than a best-of-7 format, such as that of the NBA Playoffs. That's what makes March Madness so exciting. However, it also means that just because a certain team defeated another, that result alone does not necessarily mean they are the better team; in a one-game sample, randomness, matchup quirks, and shooting variance can outweigh true team strength.

My rk	Team	KenPom rk	Diff.	Interpretation
6	St. John's (NY)	16	+10	The Red Storm won 19 of their last 20 games, including UConn by 20 points!
9	Saint Mary's (CA)	26	+17	The Gaels finished the year with a 27-6 record, including late wins and close losses to highly ranked Gonzaga and Santa Clara.
12	Virginia	18	+6	The Cavaliers made a deep run in the ACC Tournament, including barely loosing to top-seeded Duke, 74-70.
13	Florida	5	-8	My system was less convinced by Florida than KenPom was, due to their 17 points loss to Vanderbilt in the SEC Tournament.

Table 3: Selected ranking disagreements between my model and KenPom.

Saint Mary's early exit. My model ranked Saint Mary's ninth nationally, while KenPom had the Gaels 26th.[7] Saint Mary's then lost its first-round NCAA tournament game to Texas A&M, 63-50.[8] That is a clear case where my system was too optimistic.

Florida's disappointing upset. My model ranked Florida 13th, while KenPom had Florida fifth.[9] Florida was then upset by Iowa in the second round of the NCAA tournament.[10] One upset does not prove my model was "right," but it does show the value of having an independent rating. In this case, my model's recency-weighted opinion of Florida looks more cautious than KenPom's placement.

That is exactly why I built the model. The goal is not blind imitation. The goal is to create a framework that mostly tracks the best public system, but still has enough independence to generate original opinions that can later be tested against outcomes.

What these results say about my approach

Overall, the results show three things.

First, the model is **basketball-native**. It is built from possession and efficiency logic, site adjustment, opponent quality, and recency rather than generic standings data.

Second, the model is **predictively oriented**. I tuned it against margin prediction error, not against the AP poll, NCAA seeds, or subjective reputation.

Third, the model is **interpretable**. When it disagrees with KenPom, I can explain why. That

[7] KenPom ratings page, data through games of Sunday, March 22, 2026.
[8] NCAA official bracket page and Reuters game recap for Texas A&M vs. Saint Mary's, March 20, 2026.
[9] KenPom ratings page, data through games of Sunday, March 22, 2026.
[10] NCAA official tournament results and NCAA video recap of Iowa vs. Florida, March 22, 2026.

matters in analytics settings because decision-makers need more than a ranking; they need a rationale.

Conclusion

Building this metric was an exercise in both sports knowledge and modeling discipline. The final system is not presented as the secret KenPom formula. Instead, it is my best transparent approximation of the same predictive philosophy. It transforms raw NCAA box scores into a tuned, opponent-adjusted, recency-aware rating that behaves plausibly, tracks a respected benchmark, and produces testable independent views.

References

1. Ken Pomeroy, "Ratings Explanation," KenPom blog, Nov. 29, 2006.
2. Ken Pomeroy, "Ratings Methodology Update," KenPom blog, Oct. 4, 2016.
3. Ken Pomeroy, "Evidence that scoring margin matters," KenPom blog, Jan. 22, 2013.
4. Ken Pomeroy, "Mining point spread data: home court advantage," KenPom blog, May 21, 2018.
5. KenPom ratings page, data through games of Sunday, March 22, 2026.
6. NCAA.com, "2026 NCAA tournament: Bracket, schedule, scores for March Madness," updated March 23, 2026.
7. NCAA.com video recap, "Defending champs out: Iowa stuns No. 1 Florida in March Madness second round," March 22, 2026.
8. Reuters / Field Level Media, "Texas A&M cruises to wire-to-wire win vs. Saint Mary's," March 20, 2026.